

Marco López | Curriculum Vitae

✉ marco.lopez.math@gmail.com

Research Interests

My research interests are in dynamical systems, ergodic theory, fractal geometry, and computability theory. More specifically, I have done research in thermodynamic formalism, conformal iterated function systems, Hausdorff dimension of shrinking target sets, and entropy and equilibrium states for interval maps. Most recently I have been working on computability theory of dynamical systems. I am also interested in Diophantine approximation and other connections between number theory and dynamical systems.

Work Experience

2024-Present: NSF MPS-Ascend Postdoctoral Fellow, City University of New York

2023-2024: Associate Instructor, Cochise College

2021-2024: Instructor, Center for Talented Youth, Johns Hopkins University

2021-2022: Postdoctoral Researcher, Department of Mathematics, Universidade Federal do Ceará

2018-2021: Teacher-Scholar Postdoctoral Fellow, Department of Mathematics, Wake Forest University

2012-2018: Teaching Fellow, Department of Mathematics, University of North Texas

2010-2012: Graduate Teaching Assistant, Department of Mathematics and Statistics, Northern Arizona University

Education

2012–2018: Ph.D. Mathematics, University of North Texas

- Advisor: Prof. Mariusz Urbański.

- Dissertation: Hausdorff Dimension of Shrinking-Target Sets Under Non-Autonomous Systems.

2010–2012: M.S. Mathematics, Northern Arizona University

2004–2009: B.S. Mathematics, University of Arizona

Other Research Experience

AMS Math Research Communities

Participant

June 18-24, 2017

- Mentors: Dr. Vaughn Climenhaga, Dr. Daniel Thompson
- Topic: Positive entropy and hyperbolic potentials for α - β shifts.

University of Warsaw

PhD Internship

Spring 2016

- Mentor: Dr. Krzysztof Barański
- Topic: Hausdorff dimension of limit sets arising from non-autonomous systems

Teaching Experience as Instructor of Record

CUNY - The City College of New York

Instructor

Spring 2026

- Bridge to Advanced Mathematics

Cochise College

Associate Instructor

Spring 2023 - Spring 2024

- Beginning Algebra
- Precalculus Algebra
- Precalculus Trigonometry

Wake Forest University

Teacher-Scholar Postdoctoral Fellow

Fall 2018 - Spring 2021

- Discrete Dynamical Systems
- Introductory Real Analysis I
- Calculus with Analytic Geometry II
- Multivariable Calculus
- Introduction to Linear Algebra and Differential Equations

University of North Texas

Teaching Fellow

Fall 2012 - Summer 2018, Except Spring 2016

- Probability Models
- Precalculus
- Calculus II
- Survey of Mathematics with Applications
- Business Calculus
- Elementary Probability and Statistics

Northern Arizona University

Graduate Teaching Assistant

Fall 2010 - Spring 2012

- Precalculus Mathematics
- Algebra for Precalculus
- Quantitative Reasoning

Other Teaching and Mentoring Experiences

CUNY - Directed Reading Program

Mentor

2024-2025

- Mathematics for Machine Learning (Amir N., City College)
- Mathematical Finance (Allen S., Queens College)
- Invitation to Ergodic Theory (Gabriel D., Brooklyn College)
- The Mathematics of Data (Jeremy K., Baruch College)

Thrive Scholars

Instructor

Summer 2024

- Calculus 2

Center for Talented Youth, Johns Hopkins University

Online Instructor

Fall 2021 - Spring 2024

- Multivariable Calculus
- Introduction to Abstract Mathematics
- Modern Cryptography

Summer Program Instructor

- Paradoxes and Infinities

Presentations

Oral

April 2025: Learning Seminar, CUNY

- Title: Computability of MMEs

March 2025: AMS Sectional Meeting, Special Session on Ergodic Theory and Dynamical Systems celebrating Manfred Denker's 80th Birthday

- Title: Computability of MMEs for Coded Shifts

October 2024: Ergodic Theory and Dynamical Systems Seminar, CUNY

- Title: A dichotomy law for shrinking target sets under non-autonomous systems

May 2022: Dynamical Systems Seminar, Universidade Federal de Alagoas

- Title: Shrinking Target sets for nonautonomous dynamical systems

October 2021: First Iterations in Dynamics, Universidade de São Paulo

- Title: Dimensions and measures of shrinking target sets

October 2020: Math Café Colloquium, Davidson College

- Title: Fractal Geometry and Dynamical Systems

January 2020: AMS/MAA Joint Math Meetings, AMS Special Session on Fractal Geometry, Dynamical Systems, and Applications, Denver

- Title: Dimensions and measures of shrinking target sets.

November 2019: AMS Sectional Meetings, Special Session on Fractal Geometry and Dynamical Systems,

- Title: Shrinking Targets of non-autonomous systems: dimensions and measures

October 2018: AMS Sectional Meeting, Special Session on Statistical and Geometrical Properties of Dynamical Systems, San Francisco State University

- Title: Equilibrium states for (α, β) -transformations.

August 2018: Department of Mathematics and Statistics Colloquium, Wake Forest University

- Title: Hausdorff dimension of shrinking-target sets: Fractal geometry and dynamical systems

April 2018: AMS Sectional Meeting, Special Session on Quantization for Probability Distributions and Dynamical Systems, Vanderbilt University

- Title: Equilibrium states and positive entropy for α - β transformations

January 2018: Department Colloquium, Northern Arizona University

- Title: Thermodynamic formalism, dynamical systems and Hausdorff dimension

January 2018: AMS/MAA Joint Math Meetings, Special Session on Dynamical Systems: Smooth, Symbolic, and Measurable, San Diego, California

- Title: Dimension of shrinking target sets arising from nonautonomous dynamical systems

December 2017: Workshop on Fractals and Dimensions, Institut Mittag-Leffler

- Title: Dimension of shrinking targets arising from non-autonomous systems

November 2017: Dynamics Seminar, Universidad Autónoma de San Luis Potosí

- Title: Sistemas no-autónomos y blancos decrecientes

November 2017: AMS Sectional Meeting, Special Session on Random and Deterministic Dynamical Systems, University of California-Riverside

- Title: Shrinking targets and non-autonomous systems

October 2017: SACNAS National Conference, Salt Lake City

- Title: Dimension of shrinking targets from non-autonomous systems

October 2017: Max Dehn Seminar, University of Utah

- Title: Dimension of shrinking target sets arising from non-autonomous dynamics

September 2017: AMS Sectional Meeting, Special Session on Dynamics, Geometry and Number Theory, University of North Texas

- Title: Shrinking targets and non-autonomous systems

June 2016: Dynamical Systems Seminar, Institute of Mathematics of the Polish Academy of Science

- Title: Shrinking targets and non-autonomous systems

May 2016: Dynamical Systems Seminar, University of Warsaw

- Title: Non-autonomous iterated function systems and Hausdorff dimension

April 2015: Dynamical Systems Seminar, University of North Texas

- Title: Birkhoff averages of measurable rotations

April 2014: Dynamical Systems Seminar, University of North Texas

- Title: Hausdorff dimension and Bowen's formula for graph directed Markov systems

Poster.....

July 2018: Summer School in Dynamics, International Center for Theoretical Physics, Trieste

- Title: Dimension of shrinking targets from non-autonomous systems

July 2018: Resonances of complex dynamics, International Centre for Mathematical Sciences, Edinburgh

- Title: Dimension of shrinking targets from non-autonomous systems

August 2017: Conference on Current Trends in Dynamical Systems and the Mathematical Legacy of Rufus Bowen, University of British Columbia

- Title: Dimension of shrinking targets from non-autonomous systems

January 2010: AMS/MAA Joint Meetings, San Francisco, CA

- Title: Spectral characterization of Von Neumann's iterative algorithm in $\mathbb{R}^n$

October 2009: SACNAS National Conference, Dallas, TX

- Title: Spectral characterization of Von Neumann's iterative algorithm in $\mathbb{R}^n$

October 2006: SACNAS National Conference, Tampa Bay, FL

- Title: Tikhonov regularization and Bayesian statistics in inverse problems

August 2006: Young Mathematicians Conference, Ohio State University

- Title: Tikhonov regularization and Bayesian statistics in inverse problems

Awards and Grant Support

September 2024 - August 2027: NSF MPS-Ascend Postdoctoral Fellowship

September 2014 - August 2018: CONACYT (Mexico) doctoral scholarship

November 2017: UNT College of Science travel grant

October 2017: AMS Math Research Communities collaboration travel grant

August 2017: UNT Toulouse Graduate School travel grant

July 2017: UNT Department of Mathematics travel grant

May 2015: UNT Department of Mathematics travel grant

Publications

- *Computability of $\mathcal{G}$ -Bernoulli Measures and Measures of Maximal Entropy on Coded Shift Spaces* (with T. Kucherenko and Ch. Wolf) arXiv:2601.15548 (2026)
- *SRB Entropy of Markov Transformations* (with M. Jiang) Journal of Statistical Physics 188 (2022), No. 3
- *Unique equilibrium states for some intermediate beta transformations* (with L. Carapezza, and D. Robertson) Stochastics and Dynamics 21 (2021), no. 6
- *Shrinking targets for non-autonomous systems*. Nonlinearity 33 (2020), no. 7
- *Spectral characterization for Von Neumann's iterative algorithm in $\mathbb{R}^n$* . (with R. Joly, D. Mupasiri, and M. Newsome) Involve, Vol. 6 (2013), No. 2

Professional Development

Fall 2025 Cohort: Erdős Institute, Quant Finance Bootcamp

Spring 2024 Cohort: Erdős Institute, Data Science Bootcamp

Silver 2019 Cohort: MAA Project NExT

Service

February 2022: SIMIODE Expo workshop presentation

- Title: Modeling Epidemics with Skittles (joint with Tim Antonelli, Tova Brown, and Tingxiu Wang)

Fall 2020 - Fall 2021: Volunteer bilingual tutor for Mastery Hour

Fall 2020 - Spring 2021: Equity Committee

Fall 2019 - Spring 2020: Mathematics Curriculum Committee

Fall 2018 - Spring 2019: Graduate Admissions Committee

August 2018: Co-instructor for "Rocas, Agua y Electricidad", Clubes de Ciencia México, Oaxaca.

January 2018: Judge for the MAA Undergraduate Student Poster Session at the Joint Mathematics Meeting.

Citizenship

- United States of America
- Mexico

Languages

- Spanish (first language)
- English (fluent)
- Portuguese (advanced)

References

Research.....

Vasilis Chousionis:

- Associate Professor, University of Connecticut
- vasileios.chousionis@uconn.edu

Tamara Kucherenko:

- Professor, City University of New York
- tkucherenko@ccny.cuny.edu

Mariusz Urbański:

- Professor, University of North Texas
- mariusz.urbanski@unt.edu

Christian Wolf:

- Professor, Mississippi State University
- cwolf@math.msstate.edu